

The Collected Mathematical Papers of Arthur Cayley, Vol. III

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195. Report on the Recent Progress of Theoretical Dynamics (concluded)

(Conclusion from previous pages.)

I remark in conclusion, that the differential equations of dynamics (including in the expression, as I have done throughout the report, the generalized Lagrangian and Hamiltonian forms) constitute really one of the chief classes of differential equations occupying the attention of geometers. The greater part of what has been done with respect to the general theory of a system of differential equations is due to Jacobi, and he has also considered in particular, besides the differential equations of dynamics, the Pfaffian system of differential equations (including therein the system of differential equations which arise from any partial differential equation of the first order), and the so-called isoperimetric system of differential equations, that is, the system arising from any problem in the calculus of variations. In a systematic treatise it would be proper to commence with the general theory of a system of differential equations, and as a branch of this general theory, to consider the generalized Hamiltonian system; and in relation thereto to develop the various theorems which have a dynamical application. It would be shewn that the generalized Lagrangian form would be transformed into the Hamiltonian form, but the first-mentioned form would, I think, properly be treated as a particular case of the isoperimetric system of differential equations.

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[See [298] at the end of papers (Vol. IV.).]

2, Stone Buildings, March 28, 1856.

196. Note sur un Problème d'Analyse Indéterminée

[From the *Nouvelles Annales de Mathématiques* (Terquem and Gerono), t. XVI. (1857), pp. 163–165.]

EULERA a donné dans le Mémoire: *Regula facilis problemata Diophantea per numeros integros expedite solvendi* (*Commentt. Arith. Coll.*, t. II., p. 263) la solution générale d'un cas de l'équation indéterminée

$$ax^2 + 2bx + cy + 2y = ay^2 + 2gy + d,$$

en supposant que l'on ait la solution $x = a$, $y = b$ du nombre que

$$aa^2 + 2ba + cy + 2 = ab^2 + 2gb + d,$$

et en posant

$$a^2 = ac + 1,$$

où c est une quantité quelconque, l'équation sera satisfaite par les valeurs

$$\begin{aligned} x &= aa + \frac{ab}{2a}(a-1)\frac{f}{a} + (a-1)g, \\ y &= -bn + a + \frac{(a-1)b}{2a} + \frac{a-1}{a}(fd). \end{aligned}$$

En effet, en voit sans peine que ces valeurs donnent

$$ax^2 + 2bx + cy + 2 = a(a^2 + acb + ed)(ay^2 + 2gy + d) = 0.$$

En remplaçant à la place des coefficients a , β , γ , ξ , θ autant des nombres entiers tels que a_1^2 soit un entier positif non carré, on obtient une méthode pour trouver ente r des nombres qui s'il existe et le coefficient sont positifs ou négatifs; et si l'équation $a^2 = aN + 1$ est satisfaite par (a, a, c, q) et (a', a, c, q') , l'équation

$$ax^2 + \beta x + \gamma + \xi a' + xy + \rho yx + sxy + d, \quad (1')$$

est réduite à la forme connue.

L'équation indéterminée (2) existe dans celle-ci

$$(a, b, c, f, g, h)x, y_1^2 = 1, \quad \frac{(a, b, c, f, g, h)x, y, 1 = 0}{0}, \quad (2)$$

en supposant que la forme quadratique

$$(a, b, c, f, g, h)x, y, z)^2$$

est l'équation indéterminée (2). En effet en supposant que la forme quadratique

$$(a, b, c, f, g, h)x, y, z)^2 = 0$$

se transforme en elle-même au moyen d'une substitution linéaire quelconque, on peut supposer que cette substitution soit telle que l'on ait $x' = x = 1$ et au moment du plus $b = 0$. L'équation (3) se réduit alors à montrer que $x' = x$ aurait de plus $b = 0$. L'équation

(3) ne peut trouver par le méthode précédente. M. Hermite a donné cette méthode dans l'ordinaire,

$$\begin{aligned}\mathfrak{A} &= ba - f\gamma, & \mathfrak{B} &= pb - h\gamma, \\ \mathcal{B} &= aa - hg^2, & \mathfrak{H} &= ga - h\beta + h\beta.\end{aligned}$$

Il faut pour cela écrire

$$ax = \mathfrak{A}^2, \quad ay = \mathfrak{B}a - p,$$

où

$$\mathfrak{B} = \mathfrak{A}^2 + \mathfrak{B}^2 + \mathfrak{H}^2.$$

ces équations sont identiquement satisfaites par

$$(a, b, c, f, g, h)x, y, z)^2, \quad (\mathfrak{B}, b, c, f, g, h)x, y, z)^2,$$

et un moyen de cette équation et de ces valeurs

$$x' = \mathfrak{B} - \mathfrak{A}, \quad y' = \mathfrak{B}a + p,$$

on forme tout de suite équation et les valeurs

$$x = \mathfrak{B}\mathfrak{A} - p\mathfrak{B}, \quad y = \mathfrak{B}\mathfrak{A} - p\mathfrak{B} - p,$$

en forme telles que sur les trois équations

$pa = 0, \quad \mathfrak{B} = 0$ en ajoutant, on obtient $\mathfrak{A} = h\mathfrak{B}$, ce qui donne $x' = a$ et de là $x' = a$.

Cela étant, les deux équations donnent, en remplaçant $\mathfrak{B}$ par sa valeur,

$$\alpha\mathfrak{A}^2 + (h - g\mathfrak{Y}) + a + g\mathfrak{B}^2 + a\mathfrak{B} = -\beta,$$

$$(h + g\mathfrak{Y})^2 - \alpha\mathfrak{B}^2 + a(g - g\mathfrak{X})^2 = -\beta,$$

et en remettant pour $\mathfrak{A}, \mathfrak{B}, \mathfrak{Y}, \mathfrak{X}$ leurs valeurs

(page continues with formulas 4, 5, 6, 7 and conclusion)

London, 10 Mars, 1857.

197. Note on the Theory of Logarithms

[From the *Philosophical Magazine*, vol. XI. (1856), pp. 275–280.]

An imaginary quantity $x + yi$ may always be expressed in the form

$$x + yi = r(\cos \theta + i \sin \theta) = re^{i\theta},$$

where r is positive, and θ is included between the limits $-\pi$ and $+\pi$. We have in fact

$$r = \sqrt{x^2 + y^2},$$

and when x is positive,

$$\theta = \tan^{-1} \frac{y}{x},$$

but when x is negative,

$$\theta = \tan^{-1} \frac{y}{x} \pm \pi,$$

where $\tan^{-1}$ denotes an arc between the limits $-\frac{1}{2}\pi$, $+\frac{1}{2}\pi$, and where the upper or under sign is to be employed according as y is positive or negative. I use for convenience the mark $\pm$ to denote identity of sign; we may then write

$$\theta = \tan^{-1} \frac{y}{x} \pm \epsilon\pi,$$

where

$$x = +, \quad \epsilon = 0; \quad x = -, \quad \epsilon = \pm 1 \pm y.$$

It should be remarked that θ has a unique value except in the single case $x = -$, $y = 0$, where θ is indeterminately $\pm\pi$. We have, in fact, $\theta = +\pi$ or $\theta = -\pi$ according as it is considered to the limit of $x + yi$, or $x - yi$, y being positive or negative, in such an arc is the case so $r + yi$ is to be regarded.

The preceding definition is, in fact, in the case of a positive, that given by M. Cauchy in the *Exercices de Mathématique*, vol. i. [1826]; and he has there shewn that x , x' , $xx' - yy'$ being all of them positive, the above-mentioned expression reduces itself to zero. The general definition is that given to my *Mémoire sur quelques Formules du Calcul Intégral*, Liouville, vol. XII. [1847], p. 231 [40], but I was wrong in assuming that the expression always reduced itself to zero. We have, in fact, in general

$$\tan^{-1} a + \tan^{-1} \beta = \tan^{-1} \frac{a + \beta}{1 - a\beta},$$

when $1 - a\beta$ is positive; but when $1 - a\beta$ is negative (which implies that a , β have the same sign), then

$$\tan^{-1} a + \tan^{-1} \beta = \tan^{-1} \frac{a + \beta}{1 - a\beta} \pm \pi,$$

where the upper or under sign is to be employed according as a and β are positive or negative; or what is the same thing,

$$\tan^{-1} a + \tan^{-1} \beta = \tan^{-1} \frac{a + \beta}{1 - a\beta} \pm \epsilon\pi,$$

where

$$1 - a\beta = +, \quad \epsilon = 0; \quad 1 - a\beta = -, \quad \epsilon = \pm 1 \pm a = \pm 1 \pm \beta.$$

This being premised, then writing

$$\begin{aligned} \log(x + yi) &= \log \sqrt{x^2 + y^2} + i \left(\tan^{-1} \frac{y}{x} \pm \epsilon \pi \right), \\ \log(x' + y'i) &= \log \sqrt{x'^2 + y'^2} + i \left(\tan^{-1} \frac{y'}{x'} \pm \epsilon' \pi \right), \\ \log(x + yi)(x' + y'i) &= \log[(xx' - yy') + i(xy' + x'y)] \\ &= \log \sqrt{(xx' - yy')^2 + (xy' + x'y)^2} + i \left(\tan^{-1} \frac{xy' + x'y}{xx' - yy'} \pm \epsilon'' \pi \right), \end{aligned}$$

we find

$$\log(x + yi) + \log(x' + y'i) - \log(x + yi)(x' + y'i) = (\epsilon + \epsilon' - \epsilon'')\pi i.$$

Hence, considering the different cases:

I. $x = +, x' = +, xx' - yy' = +,$

$$\begin{aligned} \epsilon &= 0, \\ \epsilon' &= 0, \\ \epsilon'' &= 0, \end{aligned}$$

and therefore

$$\epsilon + \epsilon' - \epsilon'' = 0.$$

II. $x = +, x' = +, xx' - yy' = -,$

$$\begin{aligned} \epsilon &= 0, \\ \epsilon' &= 0, \\ \epsilon'' &= \pm 1 \pm (xy' + x'y) = - \left(\frac{y}{x} + \frac{y'}{x'} \right), \end{aligned}$$

and therefore

$$\epsilon + \epsilon' - \epsilon'' = \pm 1 \equiv \left(\frac{y}{x} + \frac{y'}{x'} \right).$$

III. $x = +, x' = -, xx' - yy' = +,$

$$\begin{aligned} \epsilon &= 0, \\ \epsilon' &= \pm 1 \pm y' = -\frac{y'}{x'} = -\frac{y}{x'}, \\ \epsilon'' &= 0, \\ \epsilon'' &= \pm 1 \equiv \left(-\frac{y'}{x'} \right), \end{aligned}$$

and therefore

$$\epsilon + \epsilon' - \epsilon'' = 0.$$

IV. $x = +, x' = -, xx' - yy' = -,$

$$\epsilon = 0,$$

$$\epsilon' = \pm 1 \pm y' = -\frac{y'}{x'},$$

$$\epsilon'' = \pm 1 \pm (xy' + x'y) = -\left(\frac{y}{x} + \frac{y'}{x'}\right),$$

and therefore

$$\epsilon + \epsilon' - \epsilon'' = 0 \quad \text{if } \frac{y'}{x'} = -\frac{y}{x},$$

but

$$\epsilon + \epsilon' - \epsilon'' = \pm 2 \quad \text{if } \frac{y'}{x'} = -\left(\frac{y}{x} + \frac{y'}{x'}\right).$$

V. $x = -, x' = +, xx' - yy' = +,$

$$\epsilon = \pm 1 \pm y = -\frac{y}{x} = -\left(\frac{y}{x}\right),$$

$$\epsilon' = 0,$$

$$\epsilon'' = 0,$$

and therefore

$$\epsilon + \epsilon' - \epsilon'' = 0.$$

VI. $x = -, x' = +, xx' - yy' = -,$

$$\epsilon = \pm 1 \pm y = -\frac{y}{x},$$

$$\epsilon' = 0,$$

$$\epsilon'' = \pm 1 \pm (xy' + x'y) = -\left(\frac{y}{x} + \frac{y'}{x'}\right),$$

and therefore

$$\epsilon + \epsilon' - \epsilon'' = 0 \quad \text{if } \frac{y}{x} = -\frac{y'}{x'},$$

but

$$\epsilon + \epsilon' - \epsilon'' = \pm 2 \quad \text{if } \frac{y}{x} = -\left(\frac{y}{x} + \frac{y'}{x'}\right).$$

VII. $x = -, x' = -, xx' - yy' = +,$

$$\epsilon = \pm 1 \pm y = -\frac{y}{x},$$

$$\epsilon' = \pm 1 \pm y' = -\frac{y'}{x'},$$

$$\epsilon'' = 0,$$

and therefore

$$\epsilon + \epsilon' - \epsilon'' = 0 \quad \text{if } \frac{y}{x} = -\frac{y'}{x'},$$

but

$$\epsilon + \epsilon' - \epsilon'' = \pm 2 \quad \text{if } \frac{y}{x} = \frac{y'}{x'}.$$

VIII. $x = -, x' = -, xx' - yy' = -,$

$$\epsilon = \pm 1 \pm y = -\frac{y}{x},$$

$$\epsilon' = \pm 1 \pm y' = -\frac{y'}{x'},$$

$$\epsilon'' = \pm 1 \pm (xy' + x'y) = -\left(\frac{y}{x} + \frac{y'}{x'}\right),$$

and therefore

$$\epsilon + \epsilon' - \epsilon'' = \pm 1 \quad \equiv -\left(\frac{y}{x} + \frac{y'}{x'}\right).$$

Hence writing

$$\log(x + yi) + \log(x' + y'i) - \log(x + yi)(x' + y'i) = E\pi i,$$

we have $E = 0$, except in the following cases, viz.

1. (See IV.) $x = +, x' = -, xx' - yy' = -, \frac{y'}{x'} = -\left(\frac{y}{x} + \frac{y'}{x'}\right),$ where

$$E = \pm 2 \pm \left(\frac{y}{x} + \frac{y'}{x'}\right).$$

2. (See VI.) $x = -, x' = +, xx' - yy' = -, \frac{y}{x} = -\left(\frac{y}{x} + \frac{y'}{x'}\right),$ where

$$E = \pm 2 \pm \left(\frac{y}{x} + \frac{y'}{x'}\right).$$

3. (See VII.) $x = -, x' = -, xx' - yy' = +, \frac{y}{x} = \frac{y'}{x'},$ where

$$E = \pm 2 \pm \frac{y}{x}.$$

4. (See VIII.) $x = -, x' = -, xx' - yy' = -,$ where

$$E = \pm 2 \mp \left(\frac{y}{x} + \frac{y'}{x'}\right).$$

It thus appears that when any one of the arguments $x + yi, x' + y'i$ are real and negative, or in the imaginary cases when the product of the proper expression for the arguments must be positive but their sums must be negative, then the equation

$$\log(x + yi) + \log(x' + y'i) = \log(x + yi)(x' + y'i)$$

ceases to hold in any case.

The preceding results do not apply to the case where any one of the arguments $x + yi, x' + y'i, (x + yi)(x' + y'i)$ is real and negative, in which case the formula $\log z = +\pi i$ being still negative, has in addition to its arbitrary positive or negative value $\pm\pi i$, what is the same thing the logarithm is absolutely indeterminate. The general methods are illustrated by applications to the problem of three points of rotation; the former problem is specially discussed in section 7., but the results obtained (and which, to the author remarks, afford an example of the so-called “elimination of the nodes”) do not come within the plan of the present report.

2, Stone Buildings, March 18, 1856.

198. Note upon a Result of Elimination

[From the *Philosophical Magazine*, vol. XI. (1856), pp. 378–379.]

IF the quadratic function

$$(a, b, c, f, g, h)(x, y, z)^2$$

break up into factors, then representing one of these factors by $\xi x + \eta y + \zeta z$, and taking any arbitrary quantities a, β, γ , the factor in question, and therefore the quadratic function is reduced to zero by substituting $\beta\zeta - a\eta$, $\gamma\xi - a\zeta$, $a\eta - \beta\xi$ in the place of the ideas, a more general; the function then results of the operation of performed on U , and QPU denotes the result of the operation Q performed on PU , and generally in such combinations of symbols, each operation is considered as affecting the operand denoted by means of all the symbols on the right of the operation in question. Now considering the operation QPU it is easy to see that we may write

$$QPU = (Q + P)U + (QP)U,$$

where on the right-hand side $(Q + P)$ and (QP) signify as follows: viz. $Q + P$ denotes the new algebraical product of Q and P , while QP (consistently with the general notation as before explained) denotes the result of the operation Q performed upon P as operand; and the two parts $(Q + P)U$ and $(QP)U$ denote respectively the results of the operations $(Q + P)$ and (QP) performed each of them upon U as operand. It is proper to remark that $(Q + P)$ and $(P + Q)$ have precisely the same meaning, and the symbol may be written in either form indifferently. But without a more convenient notation, it would be difficult to find the corresponding expression for $RQPU$, &c. This, however, has at one effected by means of the analytical forms called *trees* (see figs. 1, 2, 3), which contain all the trees which can be formed with one branch, two branches, and three branches respectively.

[Figures of trees with single branches and combinations are described in the text.]

The coefficients a, b, c, d are linear functions of coordinates, the equation

$$aa - b^2 = 0, \quad bd - cd = 0, \quad bd - c^2 = 0$$

(equivalent, of course, to two equations) belong to a curve of the third order in space; the edge of regression of the developable surface obtained by putting the discriminant equal to zero, or which has for its apolar

$$-x^3d^2 + 6abcd - 4ac^3 - 4b^3d + 3b^2c^2 = 0,$$

and, moreover, the above forms are the general representations of any curve of the third order, and developable surface of the fourth order. The question arises, to find the equation of the cone of the third order having an arbitrary point for its vertex, and passing through the curve of the third order. This may be done by Joachimsthal's method (vide. p. 21, q., f. be the values of a, b, c, d at the vertex of the cone, and is the equations of the cone, say in

$$aa - b^2 = 0, \quad bd - c^2 = 0,$$

for a, b, c, d write $aa + ka$, $ab + kb$, $ac + kc$, $ad + kd$; if then the equations so obtained in k are eliminated, the resulting equation will be that of the cone of the third order.

2, *Stone Buildings*, May 1, 1856.

199. Note on the Theory of Elliptic Motion

[From the *Philosophical Magazine*, vol. XI. (1856), pp. 425–428.]

If, as usual, r , $\dot{r}$ denote the radius vector and longitude, and μ the central mass, then the Vis Viva and Force function are respectively

$$T = \frac{1}{2}(\dot{r}^2 + r^2\dot{\theta}^2),$$

$$U = \frac{\mu}{r},$$

and writing

$$\frac{dT}{d\dot{r}} = p, \quad r^2\dot{\theta} = p,$$

$$\frac{dT}{d\dot{\theta}} = r^2\dot{\theta} = q,$$

we have $\dot{r}^2 = p$, $\dot{\theta}^2 = q$, and $T = \frac{1}{2}\left(p^2 + \frac{q^2}{r^2}\right)$, whence, putting $H = T - U$, the value of H is

$$H = \frac{1}{2}\left(p^2 + \frac{q^2}{r^2}\right) - \frac{\mu}{r};$$

and by Sir W. R. Hamilton's theory, the equations of motion are

$$\frac{dr}{dt} = \frac{dH}{dp}, \quad \frac{d\theta}{dt} = \frac{dH}{dq},$$

$$\frac{dp}{dt} = -\frac{dH}{dr}, \quad \frac{dq}{dt} = -\frac{dH}{d\theta},$$

or substituting for H its value, the equations of motion are

$$\frac{dr}{dt} = p,$$

$$\frac{d\theta}{dt} = \frac{q}{r^2},$$

$$\frac{dp}{dt} = \frac{q^2}{r^3} - \frac{\mu}{r^2},$$

$$\frac{dq}{dt} = 0.$$

Putting, as usual, $p = a^2(1 - e^2)$, and introducing the eccentric anomaly u , which is given as a function of t by means of the equation

$$nt + \xi = u - e \sin u,$$

(so that $\frac{du}{dt} = \frac{n}{1 - e \cos u}$), the integral equations are

$$q = na^2\sqrt{1 - e^2},$$

$$p = \frac{ena \sin u}{1 - e \cos u},$$

$$r = a(1 - e \cos u),$$

$$\theta = \varpi + \tan^{-1} \left(\frac{\sqrt{1 - e^2} \sin u}{\cos u - e} \right),$$

where the constants of integration a , e , ϖ denote as usual the mean distance, the eccentricity, the mean anomaly at epoch, and the longitude of perisaster.

Suppose that q_0 , p_0 , r_0 , θ_0 , u_0 correspond to the time t_0 (q is constant, so that $q_0 = q$), and write

$$V = nt'(u - u_0 + e \sin u - e \sin u_0),$$

joining to this the equations

$$r = a(1 - e \cos u), \quad r_0 = a(1 - e \cos u_0),$$

$$\theta - \varpi = \tan^{-1} \left(\frac{\sqrt{1 - e^2} \sin u}{\cos u - e} \right) - \tan^{-1} \left(\frac{\sqrt{1 - e^2} \sin u_0}{\cos u_0 - e} \right),$$

u , u_0 , θ will be functions of a , e , r , r_0 , θ , and consequently ξ being throughout considered as a function of a , e , ξ , H , in this view, the characteristic function V of Sir W. R. Hamilton, and according to his theory we ought to have

$$dV = \frac{1}{2}n(u - u_0)da + pdr - p_0dr_0 + qd\theta - q_0d\theta_0.$$

To verify this, I form the equation

$$\begin{aligned} dV &= [nu(u - u_0 + e \sin u - e \sin u_0) da \\ &\quad + nt' \{ (1 - e \cos u) du - (1 - e \cos u_0) du_0 - \sin u de + \sin u_0 de \}] \\ &= \frac{n \sin u}{1 - e \cos u} [de - (1 - e \cos u) da - r \sin u du + a \sin u_0 de] \\ &\quad + na\sqrt{1 - e^2} \left[d\theta - \frac{1}{\sqrt{1 - e^2}(1 - e \cos u)} ((1 - e^2) du + ae \sin u de) \right. \\ &\quad \left. - dH + \frac{1}{\sqrt{1 - e^2}(1 - e \cos u_0)} ((1 - e^2) du_0 + ae \sin u_0 de) \right], \end{aligned}$$

the coefficient of da on the right-hand side is

$$nu'(1 + e \cos u) - \frac{nut' \sin^2 u}{1 - e \cos u} - \frac{ane^2(1 - e^2)}{1 - e \cos u},$$

which vanishes, and similarly the coefficient of du , also vanishes; the coefficient of de is the difference of two parts, the first of which is

$$\frac{ne \sin u}{1 - e \cos u} de - na \sin u e^2 du,$$

and the second is the form of the form u , by virtue of which the coefficient therefore is

$$\frac{1}{2}n(u - u_0 + e \sin u - e \sin u_0).$$

We have therefore

$$\begin{aligned} dV &= \frac{1}{2}n(u - u_0 + e \sin u - e \sin u_0) da \\ &\quad + \frac{ne \sin u}{1 - e \cos u} dr - na\sqrt{1 - e^2} d\theta, \\ &\quad - \frac{ne \sin u_0}{1 - e \cos u_0} dr_0 - na\sqrt{1 - e^2} d\theta_0, \end{aligned}$$

or what is the same thing,

$$dV = \frac{1}{2}n(u - u_0) da + p dr - p d\theta dr_0 - q d\theta_0 - q d\theta_0,$$

the equation which was to be verified.

2, Stone Buildings, March 26, 1856.

200. On the Cones which pass through a given curve of the third order in space

[From the *Philosophical Magazine*, vol. XII. (1856), pp. 20–22.]

THE following investigation is connected with the theory of the cubic

$$(a, b, c, d)(x, y)^3,$$

and in particular with a theorem that the determinant formed with the second differential coefficients of the discriminant gives the square of the discriminant.

Consider the coefficients a, b, c, d as linear functions of coordinates, the equations

$$ac - b^2 = 0, \quad bd - cd = 0, \quad bd - c^2 = 0$$

(equivalent, of course, to two equations) belong to a curve of the third order in space; the edge of regression of the developable surface obtained by putting the discriminant equal to zero, or which has for its apolar

$$-a^2d^2 + 6abcd - 4ac^3 - 4b^3d + 3b^2c^2 = 0,$$

and moreover, the above forms are the general representations of any curve of the third order, and developable surface of the fourth order. The question arises, to find the equation of the cone of the third order having an arbitrary point for its vertex, and passing through the curve of the third order. This may be done by Joachimsthal's method: let a, b, c, d be the values of a, b, c, d at the vertex of the cone, and in the equations of the cone, viz.

$$ac - b^2 = 0, \quad ad - bc = 0, \quad bd - c^2 = 0,$$

for a, b, c, d write $a + ka, b + kb, c + kc, d + kd$; if then the equations so obtained in k are eliminated, the resulting equation will be that of the cone of the third order.

The substitution in question gives

$$La + Ma = R + S'a + 0,$$

$$La' + 2Ma = M'a + 0,$$

where

$$\begin{aligned} L &= \frac{1}{2}(ac - b^2), & M &= -ac + bd - 2hd, & N &= \frac{1}{2}(by - 8'), \\ L &= \frac{1}{2}(ad - c), & M &= bd - bd - cd, & N &= \frac{1}{2}(ed - dc), \\ L'' &= \frac{1}{2}3M''M + 2g', & M'' &= bd - bd + cd - 3M'', & X'' &= \frac{1}{2}(cd - bb); \end{aligned}$$

the result of the elimination is

$$4(LX' - M)(LX'' - 2M'') - (LX'' + LX' - 2MM'')^2 = 0,$$

and we have

$$LX - M^2 = 4(a - b)(c - bd) - (4 - bc)^2 = ac + 3b'(a' - b'),$$

$$LX'' - M'' = 4(a - d)(bd - ce) - (ad - bc)^2 = (ad - bc)^2,$$

$$LX'' + LX' - 2MM'' = 8(ad - bc)(bd - cd) + (4 - bd)(ad - bc) + 8M.M,$$

Write for shortness

$$by - cf = i, \quad cn - dy = m, \quad cd - bn = n, \quad ad - cf = w,$$

$$ad - by = w, \quad bd - cf = u, \quad sf - by = w, \quad dd - 2f = y,$$

values which (if a, b, c, d are linear functions of coordinates x, y, z, w) might, u, v, w this is equal to

$$-16(P - cf - syh + day - 4)bw + uw^2,$$

or, substituting and reducing,

$$P - cyh + lay = 1bw - 1ux + b'yw - 3a'(P_x - dyx) + 3y(d'a - 4'8)bd,$$

$$+k(P - 4yk) + 3by^2 + by^2 + bd - (a8)k\Omega_b'^2 + 8y'd + 4cd + 8cd,$$

I do not however refer the result obtained.

Now putting

$$bd - c^2 = p, \quad bc - cd = q, \quad ac - b^2 = r,$$

and in like manner

$$8d - q' = D, \quad By - cb = Q, \quad ay - bg = R,$$

and reducing, the final result may be expressed in the form

$$\begin{aligned} P[&= 2Dpq + (pn - 2)q + p(2a - py)r] \\ +Q[&- Wn + ((ph - 2)r - py) \quad He] \\ +R[&+ (Dy - 2(b)p - (Db - Ny) + 0] \\ &= 0, \end{aligned}$$

where a, b, c, d are current coordinates, and p, q, r are quadratic functions of a, b, c, d , the equation $P = 0$ of the second order in which the vertex (current point) of the cone (p, q, r) cone is the quadratic functions of a, b, c, d , the equation of the cone is in p, q, r, R, r are quadratic; and note VII this is equal to (suppose) is

$$F = a + ay + aav = R,$$

of the cone in question (so an arbitrary quantity), is

$$p + uq + sP'r = 0,$$

or so all length quantity

$$(bd - c^2) + (a + xc)(c - cd) + 8c^2(ac - b^2) = 0,$$

in fact this equation is evidently that of a surface of the second order passing through the curve; and there is no difficulty in shewing that it is a cone.

2, Stone Buildings, May 1, 1856.

201. Second Note on the Theory of Logarithms

[From the *Philosophical Magazine*, vol. XII. (1856), pp. 354–363.]

THE theory of logarithms, as developed in my first note (*Phil. Mag.* April 1856), [197] may be exhibited in a clearer light by considering, instead of $\log a + \log b - \log ab = E\pi i$, the equation $\log \frac{x}{a} = \log b + \log a - E\pi i$, in which more readily enables the answering *à priori* for the discontinuity in the values of E . Writing then for b the complex values $x + yi$, $c + yi$, we have

$$\log \frac{x' + y'i}{x + yi} = \log(x' + y'i) - \log(x + yi) + E\pi i,$$

where, according to the assumed definition of a logarithm,

$$\log(x + yi) = \log \sqrt{x^2 + y^2} + i \left(\tan^{-1} \frac{y}{x} \pm \epsilon\pi \right),$$

in which $\tan^{-1}$ denote an arc between the limits $-\frac{1}{2}\pi$, $+\frac{1}{2}\pi$, and $\tan^{-1} \frac{y}{x}$ is an arc between the limits $-\frac{1}{2}\pi$, $+\frac{1}{2}\pi$. The coefficient ϵ is equal to zero when x is positive; but when x is negative, then $\epsilon = +1$ or -1 , according as y is positive or negative, i.e. we have

$$\begin{aligned} x = +, & & \epsilon = 0, \\ x = -, & & \epsilon = \pm 1 \pm y; \end{aligned}$$

and of course the other logarithm in the equation have an analogous signification.

Hence, attending to the equation

$$\tan^{-1} a + \tan^{-1} \alpha = \tan^{-1} \frac{\beta + a}{1 + a\beta} \pm \epsilon''\pi,$$

where, when $1 + a\beta$ is positive, ϵ'' is equal to zero; but when $1 + a\beta$ is negative, ϵ'' is equal to $+1$ or -1 , according as $\beta + a$ is positive or negative; or what is the same thing, (β being of opposite signs when $1 + a\beta$ is negative), or to

$$\begin{aligned} 1 + a\beta = +, & & \epsilon'' = 0; \\ 1 + a\beta = -, & & \epsilon'' = \pm 1 \pm a = \mp 1 \mp \beta. \end{aligned}$$

we find

$$E = \epsilon + \epsilon' - \epsilon'',$$

where ϵ , ϵ' , ϵ'' are defined by the conditions

$$\begin{aligned} x = +, & & \epsilon = 0, \\ x = -, & & \epsilon = \pm 1 \pm y, \\ x' = +, & & \epsilon' = 0, \\ x' = -, & & \epsilon' = \pm 1 \pm y', \\ xx' + yy' = +, & & \epsilon'' = 0, \\ xx' + yy' = -, & & \epsilon'' = \pm 1 \pm xy' = \mp 1 \mp x'y. \end{aligned}$$

Suppose, to fix the ideas, that x, y are each of them positive; we have $x = 0$, and considering the several cases:

1. $x' = +, y' = +$,

Here $xx' + yy' = +, 1 + \frac{y y'}{x x'} = +$; and consequently not only $\epsilon' = 0$, but also $\epsilon'' = 0$, $\epsilon'' = 0$, and thence $E = 0$.

2. $x' = +, y' = -$,

Here $\epsilon = 0$. Moreover, xx' being positive, and $\epsilon yy'$ and $1 + \frac{y y'}{x x'}$ will have the same sign. If they are both positive, $xx' + yy'$ is positive, and $1 + \frac{y y'}{x x'}$ is negative. then $x' = 1 \pm xy' = +(\pm \frac{y}{x} - \frac{y'}{x'})$ (since xy' is negative) $= -\frac{y}{x}$, i.e. $\epsilon'' = +1$, and $1 + \frac{y' y'}{x' x'}$ being positive, we have $\epsilon'' = 0$. Hence in each case $E = 0$.

3. $x' = -, y' = +$,

Here $\epsilon' = \pm 1 \pm y'$, i.e. $\epsilon = -1$. Also xx' and yy' being each negative, $xx' + yy'$ will be negative, and therefore

$$\epsilon'' = \pm 1 \pm xy' = -\left(\frac{y'}{x'}\right),$$

i.e. if $\frac{y'}{x'} > \frac{y}{x}$, then $\epsilon'' = -1$; but if $\frac{y'}{x'} < \frac{y}{x}$, then $\epsilon'' = +1$. $1 + \frac{y y'}{x x'}$ being positive, $\epsilon'' = 0$. Hence if $\frac{y'}{x'} > \frac{y}{x}$, $E = 1 - 1 = 0$; but if $\frac{y'}{x'} < \frac{y}{x}$, then $E = 1 + 1 = 2$.

4. $x' = -, y' = -$,

Here $x' = \pm 1 \pm y'$, i.e. $\epsilon' = -1$. Also xx' and yy' being each negative, $xx' + yy'$ will be negative, and therefore

$$\epsilon'' = \pm 1 \pm xy' = -\left(\frac{y'}{x'}\right),$$

i.e. $\epsilon'' = -1$. Hence $1 + \frac{y y'}{x x'} = -$. If $\frac{y'}{x'} > \frac{y}{x}$, $\epsilon'' = 0$. Hence in such case $E = 0$.

Consider (x, y) (x', y') as the rectangular coordinates of two points A, A' . In the case which has been considered, the point A has been taken in the positive quadrant; and the preceding discussion shews that we have always $E = 0$, except in the case in which the negative portion of the axis x . But when this happens, then if the line AA' , considered as drawn from A to A' , passes from above to below the axis of x , we have $E = +2$; but if the line AA' , considered as drawn from A to A' , passes from below to above the axis of x , then $E = -2$. So that treating the points A, A' as the geometrical representations of two values $x + yi, x' + y'i$, we have for the discontinuity in the value of the difference of the logarithms in passing in a continuous quadratic; and the preceding discussion shews that we have always $E = 0$, except in the case where the finite line AA' meets the negative portion of the axis of x , in which case we have $E = +2$ or $E = -2$. The same thing is true generally in whichever quadrant

A is situated, i.e. we have always $E = 0$, except in the case in which the finite line AA' meets the negative portion of the axis of x . But when this happens, then if the line AA' , considered as drawn from A to A' , passes from above to below the axis of x , we have $E = +2$; but if the line AA' , considered as drawn from A to A' , passes from below to above the axis of x , then $E = -2$.

It thus appears that when the values $x + yi, x' + y'i$ are exhibited by means of two points A, A' in a plane, the discontinuity in the value of E depends on the path of the line which joins the two points. So that treating the points A, A' as the geometrical representations of the values $x + yi, x' + y'i$, the difference of the logarithms is the result of integration $\int_A^{A'} \frac{dz}{z}$, the path of integration being the line from A to A' . Consider in

general the definite integral

$$\int_A^{A'} du \phi u,$$

where ϕ, ϕ' are complex numbers of the form $x+yi, x'+y'i$; and take A, A' the geometrical representations of these limits, and the variable point P as the geometrical representation of the complex variable z . The value of the definite integral will depend in general not only upon the values which ϕ, ϕ' assume as constants, but also upon the path which the variable z supposes to be successively assumed in passing from A to A' .

To order therefore to give a precise signification to the notation, we must for the path of the point P , and it is natural to assume that the path is a right line (of course there exist also an infinity of paths which give the same value to the definite integral, or as we may call them, paths equivalent to the right line); but the consideration of these would be a tedious complication of the definition, and it is better to attend to the single path—the right line). The definition in an exact sense executed into an analytical mode of computing the negative part of the axis of x in which case we have $E = +2$ or $E = -2$ (according as the line goes from above to below or below to above the axis), then there may be considered as a more general spherical mass having a double natural connection of integration without limits. (For which is in fact identical with that between ϕ and ϕ' , and in fact the equation between θ, θ' is

$$\tan \theta \tan \phi = \frac{1}{R},$$

which, if $\phi = \cos a$, gives $\theta = \sin(R - r)$.

Richelot has shewn, by precisely similar reasoning, that for circles of the sphere we have

$$\frac{d\phi}{\sqrt{\cos^2 r - (\cos R \sin u + \sin R \cos u \cos 2\phi)^2}} = \frac{d\phi'}{\sqrt{\cos^2 r - (\cos R \sin u + \sin R \cos u \cos 2\phi')^2}},$$

which is of the form

$$\frac{d\phi}{\sqrt{1 - (k + \mu \cos 2\phi)^2}} = \frac{d\phi'}{\sqrt{1 - (k + \mu \cos 2\phi')^2}},$$

where

$$k = \frac{\cos R \sin u}{\cos r}, \quad \mu = \frac{\sin R \cos u}{\cos r}.$$

2, *Stone Buildings, September 13, 1856.*

202. Supplementary Remarks on the Porism of the In-and-Circumscribed Triangle

[From the *Philosophical Magazine*, vol. XIII. (1857), pp. 19–30.]

IN my former papers (see *Phil. Mag.* August and November, 1853, [115, 116]) I established (as part of a more general one) the following theorem, viz. the condition that there may be inscribed in the conic $U = 0$ an infinity of triangles circumscribed about the conic $V = 0$, is that if we develop in ascending powers of k the square root of the discriminant of $kU + V$, the coefficient of k in this development must vanish. Thus writing

$$\text{disc.}(kV + U) = (K, \mathfrak{R}, \mathfrak{R}', K')(k, 1)^3,$$

the condition in question is found to be

$$36\mathfrak{R}^2 - 4KW = 0.$$

The following investigations, although relating only to particular cases of the theorem, are, I think, not without interest. If the equation of the conic containing the angles is

$$U = \xi ayx + \mathfrak{B}cx + \mathfrak{C}ay = 0,$$

and the equation of the conic touched by the sides is

$$V = a^2x^2 + b^2y^2 + c^2z^2 - 2bcxz - 2cayx - 2abxy = 0,$$

that is,

$$K = -A, \quad \mathfrak{R} = -\frac{1}{3}(a + b + c), \quad \mathfrak{R}' = -\frac{1}{3}(a + b + c)^2, \quad K' = \text{disc.}$$

[The detailed investigation extends through the analysis of porism conditions for the in-and-circumscribed triangle, involving determinants, conics, and the formulae of Richelot for the sphere.]

By comparing with the corresponding formula *in plano*, we arrive at Richelot's conclusion, that the formulæ for the sphere may be deduced from those *in plano* by writing in the place of $\frac{R}{r}$, $\frac{a}{r}$, the functions $\frac{\tan R}{\tan r}$, $\frac{\sin a}{\cos R \sin r}$, respectively.

2, Stone Buildings, October 1, 1856.

203. On the Theory of the Analytical Forms called Trees

[From the *Philosophical Magazine*, vol. XIII. (1857), pp. 172–176.]

A SYMBOL such as $AU_0 + BU_1 + \dots$, where A , B , &c. contain the variables x , y , &c. in respect to which the differentiations are to be performed, partakes of the natures of an operand and operator, and may be therefore called an Operandator. Let P , Q , R , &c. be any operandators, and let U be a symbol of the same kind, or to the ideas a mere operand; PU denotes the result of the operation of P performed on U , and QPU denotes the result of the operation Q performed on PU ; and generally in such combinations of symbols, each operation is considered as affecting the operand denoted by means of all the symbols on the right of the operation in question. Now considering the operation QPU , it is easy to see that we may write

$$QPU = (Q + P)U + (QP)U,$$

where on the right-hand side $(Q + P)$ and (QP) signify as follows: viz. $Q + P$ denotes the mere algebraical product of Q and P , while QP (consistently with the general notation as before explained) denotes the result of the operation Q performed upon P as operand; and the two parts $(Q + P)U$ and $(QP)U$ denote respectively the results of the operations $(Q + P)$ and (QP) performed each of them upon U as operand. It is proper to remark that $(Q + P)$ and $(P + Q)$ have precisely the same meaning, and the symbol may be written in either form indifferently. But without a more convenient notation, it would be difficult to find the corresponding expression for $RQPU$, &c. This, however, has been effected by means of the analytical forms called *trees*.

The inspection of these figures will at once shew what is meant by the term in question, and by the terms root, branches (which may be either main branches, intermediate, or free branches), and knots (which may be either the root itself, or proper knots, or the extremities of the free branches). To apply this to the question in hand, PU consists of a single term, represented by fig. 1 (*bis*); QPU consists, as we have seen, of two terms represented by the two parts of fig. 2 (*bis*); the first part represents the term $(Q + P)U$, and the second part represents the term $(QP)U$. And it is obvious then fig. 2 (*bis*) is at once formed from the figure 1 (*bis*) by adding on a branch terminated by Q at each of the knots: the upper part of fig. 1 (*bis*). In like manner $RQPU$ consists of six terms represented by the six parts of fig. 3 (*bis*), and this figure is at once formed from fig. 2 (*bis*) by adding on a branch terminated by R at each of the knots of each part of fig. 2 (*bis*). It is hardly necessary to remark that the first part of fig. 3 (*bis*) denotes what, in the notation first explained, would be denoted by $(R + Q + P)U$, the second term what would be like manner be denoted by $(RQ + P)U$, and so on; the last part being the term which would be denoted by $((RQ)P)U$, viz. the operation upon Q , giving the operandator RQ , which operates upon P , giving the operandator $(RQ)P$, which finally operates upon U .

The figures 1 (*bis*), 2 (*bis*), 3 (*bis*), contain the same trees as are contained in the corresponding figures 1, 2, 3, &c.; only, on account of the different modes of filling up, trees are considered as so many distinct trees in a figure of the second set which correspond to one and the same tree in the corresponding figure of the first set. A difference in the number of trees first occurs in the figures 3 and 3 (*bis*), the number of trees in figure 3

being 4, while the number of trees in figure 3 (*bis*) is 9. Again, on account of similarity, only the figures of the first set are considered as truly distinct from each other; thus, in fig. 3 the first tree is fill up in two way, viz. the third tree of fig. 3 (*bis*); in like manner U at the root and R, Q, P at the other knots in any one many ways; but the trees so obtained are considered as one and the same tree, for, then we must only fill up the first tree of fig. 3 (*bis*) in the order of fig. 3 (which figure for shortness in considered as one of fig. 3 (*bis*)). Again, in fig. 3, the second tree contains the latter contains six trees, viz. the first tree, the second, third and fourth tree, the fifth tree and the sixth tree of fig. 3 (*bis*) correspond respectively to the first tree, the second tree, the third tree, and the fourth tree of fig. 3, &c. To derive fig. 3 (*bis*) from fig. 3, we must fill up the trees of fig. 3 with U, R, Q, P in all the ways possible: thus the four orders of fig. 3 of with U, R, Q, P , may be filled up in one way; the second of fig. 3 (*bis*) correspond respectively to the first tree, the second tree, the third tree, and the fourth tree of fig. 3, &c. To derive fig. 3 (*bis*) from fig. 3, we must fill up the trees of fig. 3 with U, R, Q, P in all the ways possible: the only restriction is that the root to be filled in every way, but the trees so obtained are considered as one and the same tree, and there may only be no restriction in either form indifferently. The number of trees in the resulting fig. 3 (*bis*) which may then be filled in any direct manner are those in the annexed figures, and the number is therefore two. It is not difficult to see that we have in this case (R_r being the number of such trees with r free branches),

$$(1-x)^{-1}(1-x^2)^{-R_1}(1-x^3)^{-R_2}\dots = 1+x+2R_2x^2+2R_3x^3+2R_4x^4+\&c.;$$

To obtain the law for the number A_n of the trees with n branches, we consider how the trees with n branches can be formed by means of those of a smaller number of branches. A tree with n branches has either a single main branch, or else two main branches, three main branches, etc. to n main branches. If the tree has one main branch, it can only be formed by adding on to this main branch a tree with $(n-1)$ branches, i.e. A_n contains a part A_{n-1} . If the tree has two main branches, then $p+q$ being a partition of $n-2$, the tree can be formed by adding on to one main branch a tree of p branches, and to the other main branch a tree of q branches; the number of trees so obtained is A_pA_q ; this, however, assumes that the parts p and q are unequal; if they are equal, it is easy to see that the number of trees is only $\frac{1}{2}A_p(A_p+1)$. Hence $p+q$ being any partition of $n-2$, A_n contains the part A_pA_q , if p and q are unequal, and the part $\frac{1}{2}A_p(A_p+1)$ if p and q are equal. In like manner, considering the trees with three main branches, then if $p+q+r$ is any partition of $n-3$, A_n contains the part $A_pA_qA_r$ if p, q, r are unequal; the part $\frac{1}{2}A_p(A_p+1)A_r$ if p and q are equal, and so on, until lastly we have a single tree with n main branches, or A_n contains the part A_1 . A little consideration will shew that the preceding rule for the formation of the number A_n is completely expressed by the equation

$$(1-x)^{-A_1}(1-x^2)^{-A_2}(1-x^3)^{-A_3}\dots = 1+A_1x+A_2x^2+A_3x^3+A_4x^4+\&c.,$$

and consequently that we may, by means of this equation, calculate successively for the different values of the number A_n of the trees with n branches. The calculation may be effected very easily as follows: [the table as originally printed contained at the end of it

some errors of calculation which were corrected, B.A. Report for 1875, p. 258].

$A_1 =$	(1)	1	1	1	1	1	1	1	1	1	$(1-x)^{-1}$
$A_2 =$	1	(1)	2	2	3	3	4	4	5	5	$(1-x^2)^{-1}$
$A_3 =$	1	1	(2)	4	5	8	10	14	18	24	$(1-x^3)^{-2}$
$A_4 =$	1	1	2	(4)	6	10	14	22	30	44	$(1-x^4)^{-4}$
$A_5 =$	1	1	2	3	(9)	14	25	41	64	96	$(1-x^5)^{-9}$
$A_6 =$	1	1	2	3	5	(20)	29	56	95	153	$(1-x^6)^{-20}$
$A_7 =$	1	1	2	3	5	9	(48)	75	142	247	$(1-x^7)^{-48}$
$A_8 =$	1	1	2	3	5	9	20	(115)	181	348	$(1-x^8)^{-115}$
$A_9 =$	1	1	2	3	5	9	20	48	(286)	460	$(1-x^9)^{-286}$
$A_{10} =$	1	1	2	3	5	9	20	48	115	(719)	$(1-x^{10})^{-719}$
$A_n =$	1	1	2	3	5	9	20	48	115	286	719
for $n =$	1	2	3	4	5	6	7	8	9	10	11

I have had occasion, for another purpose, to consider the question of finding the number of trees with a given number of free branches, bifurcations at least. Thus, when the number of free branches is three, the trees of the form in question are those in the annexed figure, and the number is therefore two. It is not difficult to see that we have in this case (R_r being the number of such trees with r free branches),

$$(1-x)^{-1}(1-x^2)^{-R_2}(1-x^3)^{-R_3}\dots = 1+x+2R_2x^2+2R_3x^3+2R_4x^4+\&c.;$$

and a like process of development gives:

$$\begin{array}{l} R_n = \\ \text{for } n = \end{array} \left| \begin{array}{cccccc} 1 & 2 & 5 & 12 & 33 & 90 \\ 2 & 3 & 4 & 5 & 6 & 7 \end{array} \right.$$

I may mention, in conclusion, that I was led to the consideration of the foregoing theory of trees by Professor Sylvester's researches on the change of the independent variables in the differential calculus.

2, Stone Buildings, January 2, 1856.

204. On a Problem in the Partition of Numbers

[From the *Philosophical Magazine*, vol. XIII. (1857), pp. 245–248.]

It is required to find the number of partitions into a given number of parts, such that the first part is unity, and that no part is greater than twice the preceding part.

Commencing to form the partitions in question, those are

1 1 1 1 1 1 1 1 &c.;
 1 2; 1 1 2; 1 1 1 2; 1 1 1 1 2; &c.;
 1 1 2 3; 1 1 2 4; &c.;

and if we were to proceed to the 4-partitions, each 3-partition ending in 1 would give rise to two such partitions; each 3-partition ending in 2 to four such partitions; each 3-partition ending in 3 to six partitions; each 3-partition ending in 4 to eight such partitions. We form in this manner the Table:

Number of partitions	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Totals
1-partitions	1														1
2-partitions		1	1												2
3-partitions			1	2	1	1									5
4-partitions				1	2	3	3	3	2	2	1	1			18
5-partitions					1	2	3	5	6	8	10	11	10	11	100

and we are thus led to the series

1, 2, 5, 18, &c.,

where, considering it as the first term of each series, the first differences of any series are the terms twice repeated of the next preceding series: thus the differences of the fourth series are 1, 1, 2, 3, 4, 6, 6. It is moreover clear that the first half of each series is precisely the series which immediately precedes it; we need, in fact, only consider a single infinite series 1, 2, 5, 6, &c. It is to be remarked, moreover, that in the column of totals, the total of any term is precisely the first number in the next succeeding line.

Consider in general a series A, B, C, D, E , &c., and a series A', B', C', D', E' , &c. derived from it as follows:

$$\begin{aligned}
 A' &= 2A, \\
 B' &= 2B, \\
 C' &= 2A + 2B, \\
 D' &= 2A + 2B + 2C, \\
 E' &= 2A + 2B + 2C + 2D, \\
 F' &= 2A + 2B + 2C + 2D + 2E, \\
 &\text{\&c.;}
 \end{aligned}$$

viz. the first differences of the series 0, A', B', C', D', E' , &c. are A, A, B, B, C, C, D, D , &c. Multiplying by 1, a, a^2 , &c. and adding, we have

$$A' + B'a + C'a^2 + D'a^3 + E'a^4 + \text{\&c.} = 2(A + Ba + Ca^2 + \text{\&c.})$$

and if we form in a similar manner $A'', B'', C'', D'', \&c.$ from A', B', C', D' , and so on, we have

$$A'' + B''a + C''a^2 + \&c. = \frac{1+a}{1-a}(A + Ba + Ca^2 + \&c.),$$

$$A''' + B'''a + C'''a^2 + \&c. = \frac{1+a}{1-a} \cdot \frac{1+a^2}{1-a^2}(A + Ba + Ca^2 + \&c.);$$

and so on. While $A = 1$, and suppose that the process is repeated an indefinite number of times, we have

$$1 + 2a + 5a^2 + 2a^3 + \&c. = \frac{1+a}{1-a} \cdot \frac{1+a^2}{1-a^2} \cdot \frac{1+a^3}{1-a^3} \cdot \frac{1+a^4}{1-a^4} \&c.$$

and the coefficients 1, 2, 5, 18, &c. are precisely simple algorithms for the calculation of the coefficients of $a, a^2, \&c.$ which is the third notation of the following theorem:

$$1 + 2a + 5a^2 + 18a^3 + 100a^4 + \&c. = \frac{1+a}{(1+a)^2(1+a^2)} - x^{42}(1+a^2)^{-43} - x^{44}(1+a^3)^{-44} \&c..$$

which gives also to the following very simple algorithm for finding the number of coefficients of $a, a^2, \&c.$ (in question are positive numbers as is the case);

$$1 + 2a + 6a^2 + 2a^3 + \&c. ;$$

this, on multiplying by 1, $a, a^2, \&c.$ (in question are positive numbers and the doubling $3a^2, 4a^3, \&c.$ and consequently, the operation $-3a^2(1+a^2)^{-3} - 4a^3(1+a^4)^{-3} \&c.$ (we may write the operation as a series, $1 + 2a + 5a^2 + 6a^3 + 6a^4 + \&c.$), then this product is the same as the expression which represents the number of partitions of n , viz. as in the part question.

The proceeding expression for $1 + 2a + 5a^2 + 6a^3 + \&c.$ as part greater than twice the preceding part is equal to the number of partitions of $n + 1$ into parts not greater than the previous one, may by itself become to the following:

$$1 + 2a + 5a^2 + 2a^3 + 4a^4 + \&c.$$

For example, the partitions of 5, 1, 2; are with the parts 1, 1, 7; 2, 3 = 1; the number of 3-partitions is 4; and by the second part of the theorem, the number of 4-partitions is

$$3(1 + 2 + 4 + 6) = 36.$$

the number of which are 1, 2, 4. Hence, by the first part of the theorem, the number of 5-partitions is 8, and by the second part of the theorem, the number of 3-partitions is 5.

2, Stone Buildings, March 17, 1857.

205. Note on the Summation of a Certain Factorial Expression

[From the *Philosophical Magazine*, vol. XIII. (1857), pp. 419–423.]

MR KIRKMAN some months ago communicated to me a formula for the double summation of a factorial expression, to which formula he had been led by his researches on the partition of polygons. The formula in a slightly altered form is as follows: viz.

$$\sum_{m=0}^{m=n} \frac{(\rho + s + 2)^m (s)^m}{(\rho)^m} \cdot \frac{(\rho + s - 1 - 2)^{m-1}}{(s - 1)^{m-1}} = \frac{2k}{(k + 1)^{(k+1)}} (s)^{(k+1)}$$

where the summation extending from $z = 0$ to $z = n - 1$. In the particular case where $k = 0$, this is a formula well known. In the same case the series except those in which $\rho = s$ vanish; and putting therefore $k = 0$, and making a slight change in the form of the right-hand side, the formula becomes

$$\sum_{m=0}^{m=n} \frac{(2\rho + 2k)^m}{\rho^m} \cdot \frac{(2\rho + 1)^{(\rho-1)}}{(2\rho + 1)^{(\rho-1)}} = \frac{(2\rho + 1)^{(\rho-1)}}{(\rho - 1)^{(\rho-1)}}$$

the summation extending from $z = 0$ to $z = n - 1$.

We have, in the notation of Gauss, $[m]^z = m(m - 1) \dots 2.1 = \Pi m$, and a factorial $[m]^z$ is expressed in terms of the function Π by the formula $[m]^z = \Pi m \div \Pi(m - n)$. Write also

$$\Pi(m - 1) = \Pi(m) \div [m].$$

we have

$$\Pi m = m \Pi(m - 1) = m \Pi(m - 1),$$

$$\Pi(2m + 1) = 2m \Pi(2m - 1).$$

2, *Stone Buildings*, January 5, 1857. [continues on the next page set]

[End of pp. 201–250 chunk.]